Lab 3
Git & GitHub
based on slides by Gian Carlo and Chloe

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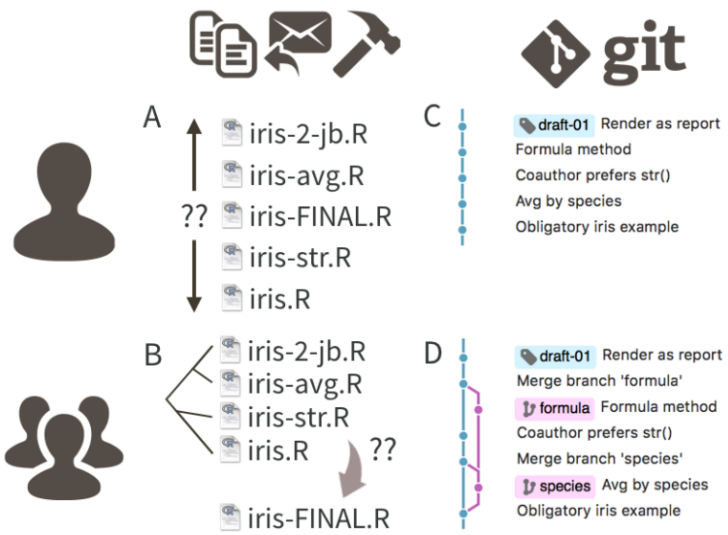
Welcome!

Today, we are learning about git and GitHub.

Version control with git and GitHub[1]

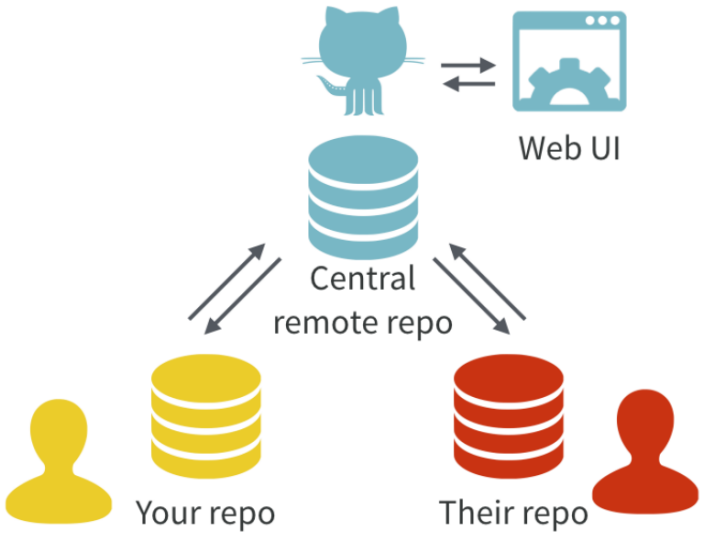
- git is a version control system that tracks and manages changes to software code



[1] Excuse me, do you have a moment to talk about version control? Bryan, J. (2017)

Cloud storing with GitHub[2]

- GitHub is a cloud-based storage site (like Google Drive)



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- Makes sharing your projects with the world super easy
- Admin control over repos and organizations
- GitHub pages to create and host websites
- There are other alternatives (e.g., Bitbucket, GitLab)

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- Other commands: `git status`, `git checkout`, `git merge`

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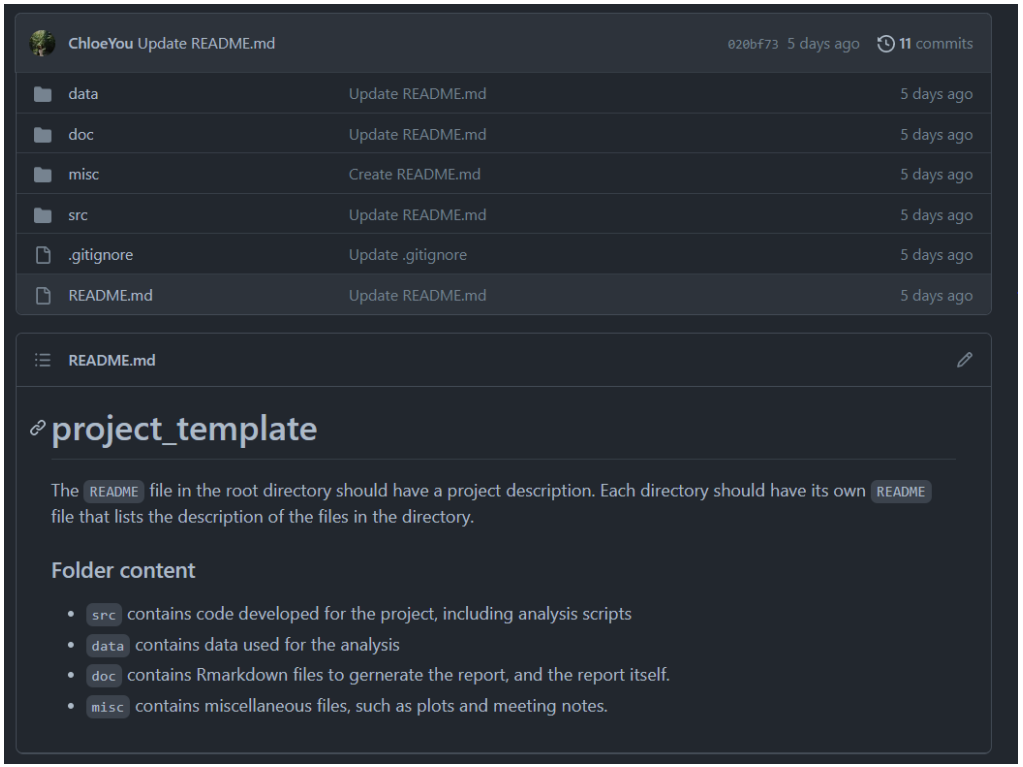
- For STAT 450, all work will be in a single repo
- **Use `README.md` files**. They make your repo readable!

How to organize your repo

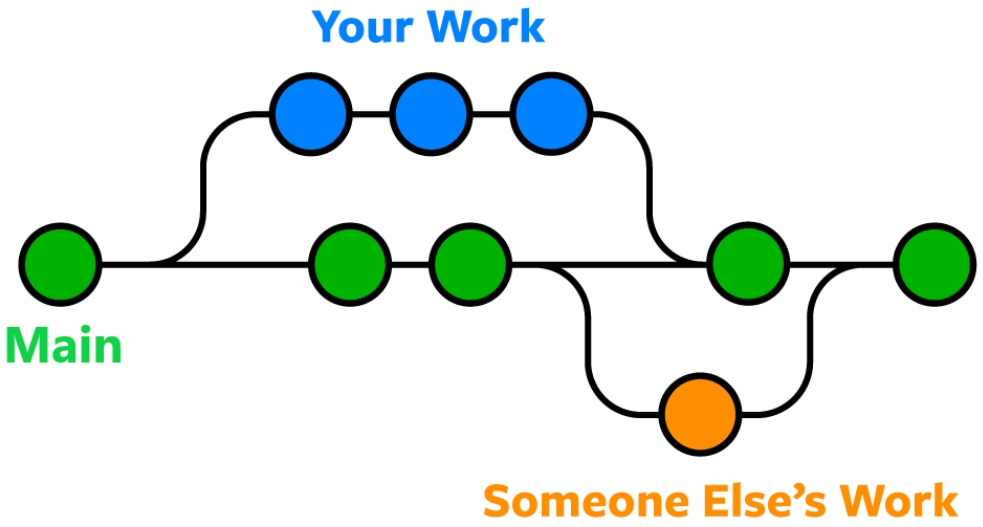
A good structure for your repo is usually:



Subdirectories (within reason) should have a `README.md` file too!

Working on branches

- Branching means you diverge from the main line of development and continue to do work without messing with that main line.[3]



[3] https://www.nobledesktop.com/learn/git/git-branches

Merge conflicts

- Conflicts generally arise when two people have changed the same lines in a file, or if one person deleted a file while another person was modifying it.[4]
- Demo: branching, resolving merge conflict, creating a pull request, some git commands

[4] https://www.atlassian.com/git/tutorials/using-branches/merge-conflicts

Now your turn!

1. Create a new branch in your project repo
2. Clone it to your computer
3. Add your client meeting notes from this morning to the `misc`
4. Stage, commit with a message, and push your changes to your branch
5. Add a description of your file in the README.md file in the `misc` directory on your branch
6. Stage, commit with a message, and push your changes to your branch
7. On Github, create a pull request to the main branch and assign one of your teammates for review.
8. Copy the pull request link and submit it to Canvas under assignments `Lab 3`.
9. Your teammate should review it and merge the branch into main. You might need to work in groups to resolve merge conflicts, but it's a good exercise!